

# Cloud Infrastructure Optimization Using K-Means Clustering

## 1. Introduction

This project applies unsupervised machine learning techniques to analyze and segment cloud infrastructure configurations for optimizing resource allocation. Using clustering methods, similar system configurations were grouped to identify workload tiers and improve efficiency.

## 2. Problem Statement

Accenture AI Labs aims to optimize cloud resource allocation by grouping similar infrastructure configurations based on CPU usage, memory, storage, network bandwidth, and energy consumption. Unsupervised learning techniques such as K-Means clustering were applied along with PCA visualization.

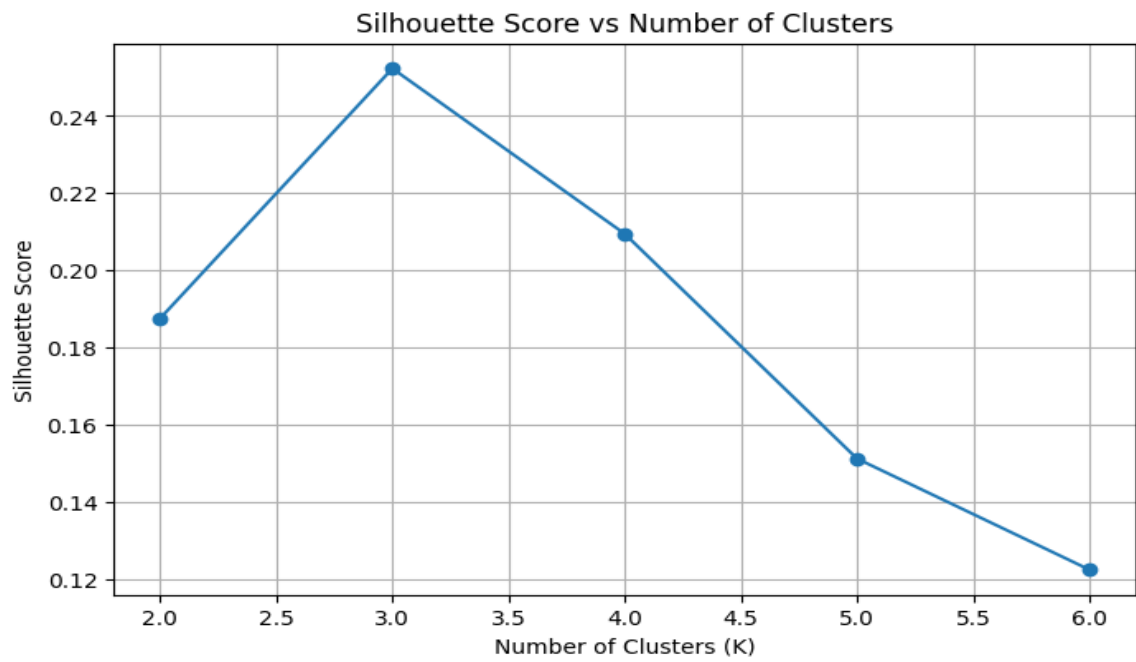
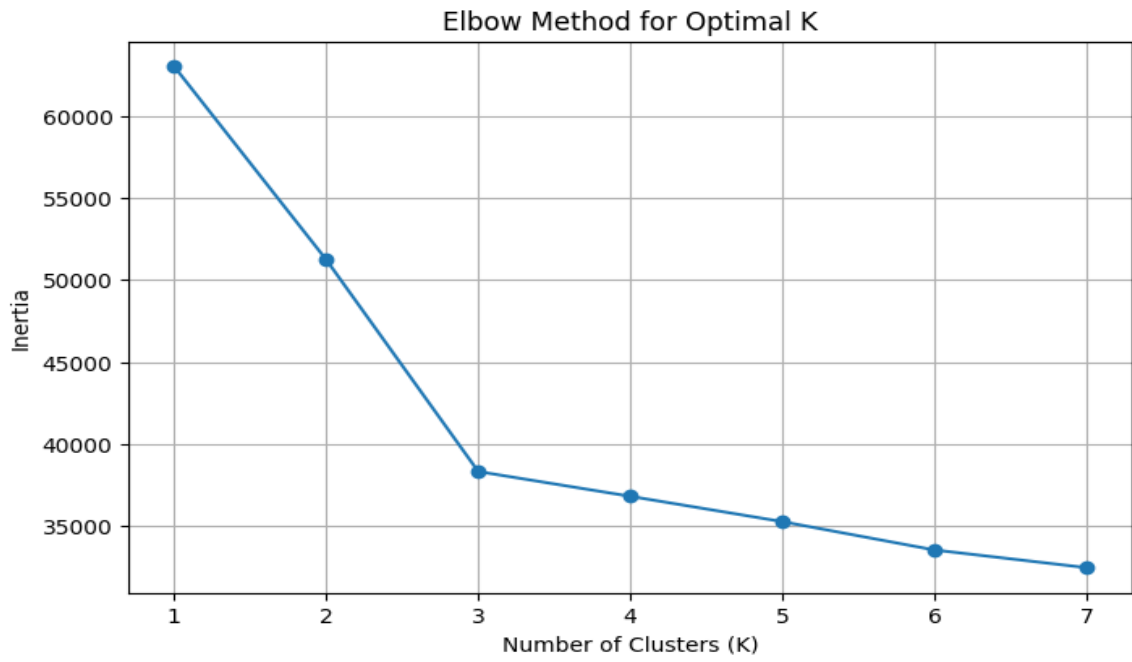
## 3. Methodology

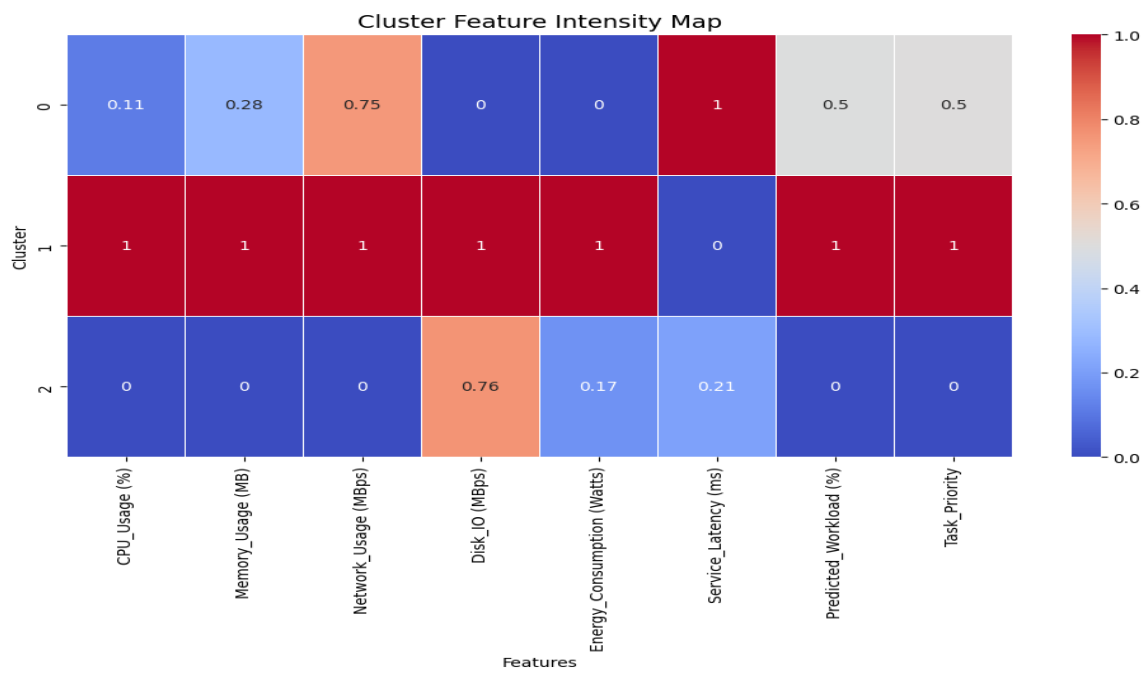
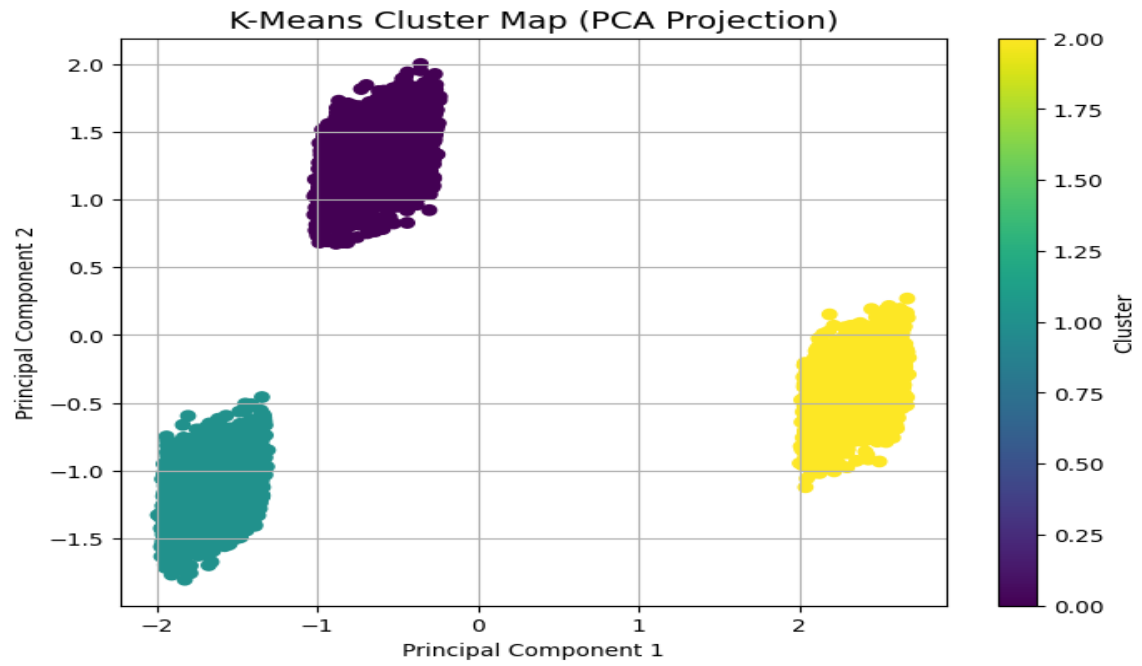
- 1 Data Cleaning and validation of missing values.
- 2 Removal of Optimized\_Resource\_Allocation to maintain unsupervised modeling.
- 3 Encoding of categorical feature Workload\_Type.
- 4 Feature scaling using StandardScaler.
- 5 Dimensionality reduction using PCA.
- 6 K-Means clustering to segment configurations.
- 7 Validation using Elbow Method and Silhouette Score.

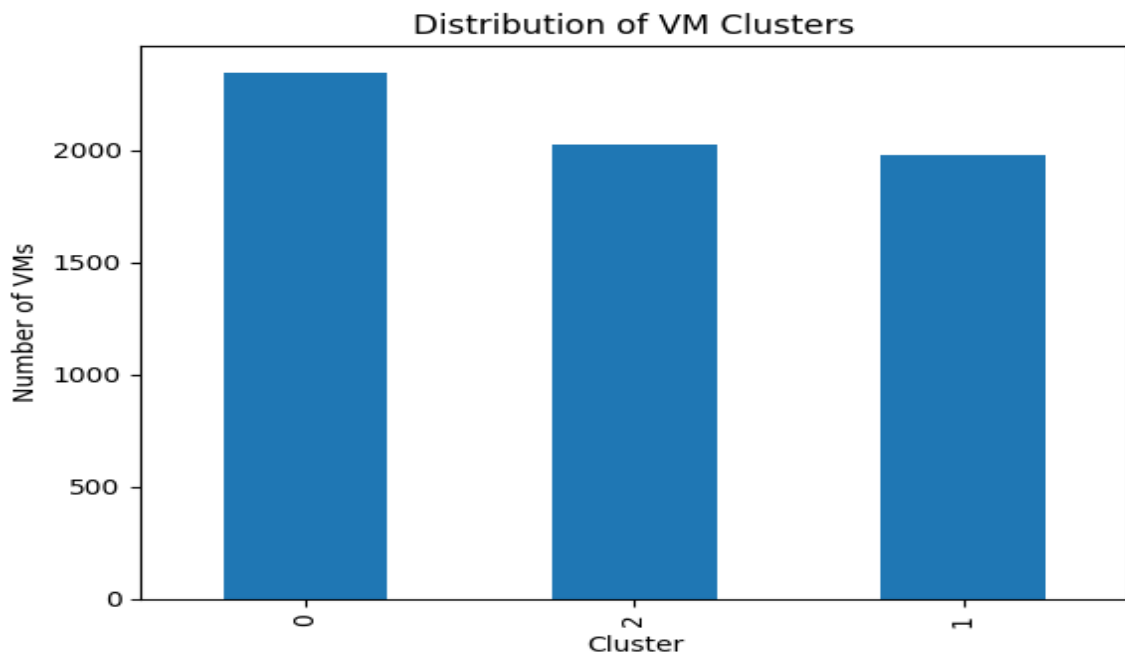
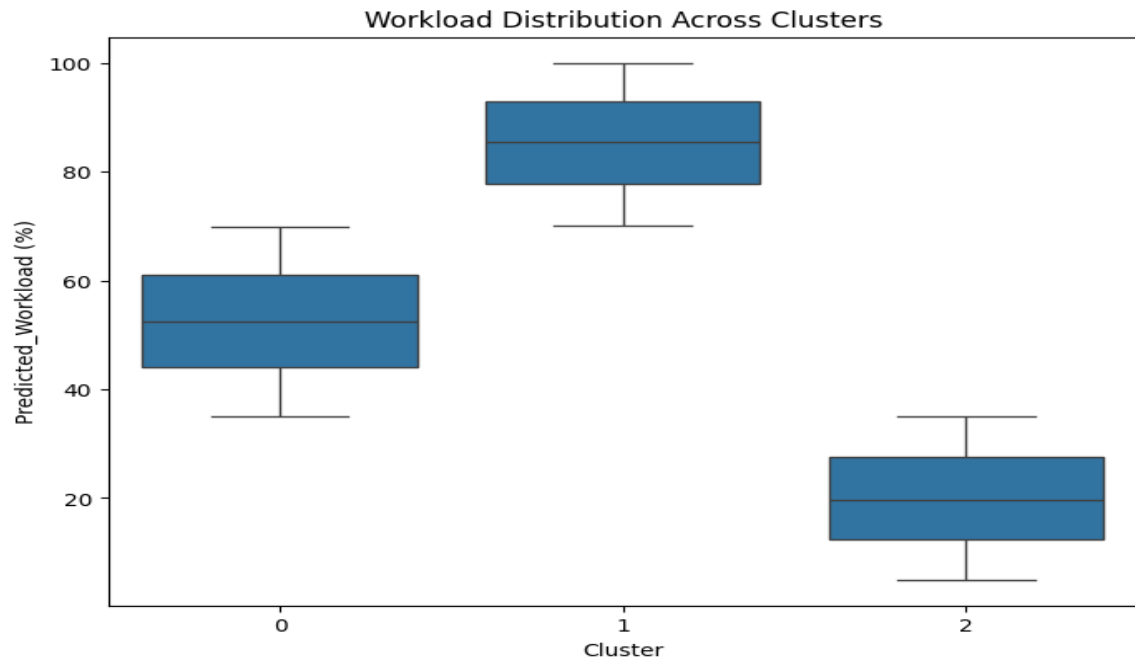
## 4. Model Evaluation

Optimal cluster count ( $K=3$ ) was determined using both Elbow Method and Silhouette Analysis. The final model achieved a Silhouette Score of approximately 0.605, indicating strong cluster separation.

## 5. Visual Analysis







## 6. Key Findings

Three distinct workload tiers were identified: High-demand systems, Balanced systems, and Low-demand systems. Workload intensity and task priority were key drivers of cluster separation. The results provide actionable insights for cloud optimization.

## 7. Optimization Recommendations

High-demand clusters require stable resource allocation and monitoring. Balanced clusters should be maintained with predictive scaling. Low-demand clusters present consolidation and energy-saving opportunities.

## **8. Conclusion**

The project successfully demonstrated how unsupervised learning and dimensionality reduction can optimize cloud infrastructure management. The strong silhouette score and visual separation confirm effective clustering aligned with the problem objective.